

Some of these problems have a closed-answer format, similar to what you might find on the final exam.

Questions marked with (*) are a bit more complex, you might want to skip them at the first read. Hints available in the final page.

3.1. Lipschitz or not. Give an example of:

- (a) A $\frac{1}{2}$ -Lipschitz function $f: [0, 1] \rightarrow [0, 1]$ that has no fixed point.
- (b) A map $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that has no fixed point, but it is an isometry, i.e.,

$$\|f(x) - f(x')\| = \|x - x'\| \text{ for all } x, x' \in \mathbb{R}^2.$$

3.2. Problems at the origin. Show that there is no continuous function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $f(x, y) = xy/(x^2 + y^2)$ for all $(x, y) \neq (0, 0)$. On the other hand, show that there is exactly one continuous function $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $g(x, y) = xy/\sqrt{x^2 + y^2}$ for all $(x, y) \neq (0, 0)$.

3.3. Open, closed, complete and compact. For each of the following subsets of $\mathbb{R}^N$ say whether they are open/closed/none/compact (with respect to the standard Euclidean structure). Try to prove “efficiently” your assertions.

1. $E_1 := \{x \in \mathbb{R}^3 : 0 < x_2 \leq 2\}$
2. $E_2 := \{x \in \mathbb{R}^n : \sin(\|x\|) \geq \frac{1}{4}\}$
3. $E_3 := \bigcup_{n \geq 1} \{x \in \mathbb{R}^3 : x_1^4 - \frac{1}{n}x_3^2 - x_2^2 > \frac{1}{n}, x_1 + x_2 < 6\}$
4. $E_4 := \{(x, y) \in (\mathbb{R}^n \times \mathbb{R}^n) : x \cdot y > \frac{1}{2}\|x\|\|y\|\}$
5. $E_5 := \{X \in \mathbb{R}^{n \times n} : \text{the matrix } X \text{ is invertible}\}$
6. $E_6 := \{X \in \mathbb{R}^{n \times n} : \text{the matrix } X \text{ is symmetric}\}$
7. $E_7 := \{X \in \mathbb{R}^{n \times n} : \text{the entries of the matrix } X \text{ are either 0 or 1}\}$
8. $E_8 := \{x \in \mathbb{R}^n : |x_i| \leq 6, \text{ for all } 1 \leq i \leq n\}$
9. $E_9 := E_8 \times E_3 \subset \mathbb{R}^{n+3}$
10. $E_{10} := E_2 \cap E_8 \subset \mathbb{R}^n$

3.4. Multiple choice. Select all the statements below that are true.

- (a) A nonempty open strict subset of $\mathbb{R}^n$ cannot be compact.
- (b) A nonempty open strict subset of $\mathbb{R}^n$ cannot be complete.
- (c) A complete subset of $\mathbb{R}^n$ contains all its accumulation points.
- (d) Countable union of complete subsets of $\mathbb{R}^n$ is complete.

- (e) A subset is closed in $\mathbb{R}^n$ if and only if it is complete as metric space itself (with the distance inherited from $\mathbb{R}^n$).

3.5. Multiple choice. Select all the statements below that are true.

- (a) If you spread a map of Zurich on your desk, then one point of the desk will coincide with its representation on the map (in an ideal world).
(b) If $f: [0, 1] \rightarrow [0, 2]$ is $\frac{1}{2}$ -Lipschitz, then f has a fixed point.
(c) If $f: [0, 2] \rightarrow [0, 1]$ is $\frac{1}{2}$ -Lipschitz, then f has a fixed point.
(d) If $f: [0, 1] \cup [2, 3] \rightarrow [0, 1] \cup [2, 3]$ is continuously differentiable with $|f'(x)| < 1$ for all $x \in [0, 1] \cup [2, 3]$, then it has a fixed point.
(e) (*) If $f: [0, 1] \rightarrow [0, 1]$ is differentiable with $|f'(x)| < 1$ for all $x \in (0, 1)$, then it has a fixed point.

3.6. Multiple choice. Select all the statements below that are true.

- (a) The function $(x, y) \mapsto x + y$ is uniformly continuous in $\mathbb{R}^2$.
(b) The function $(x, y) \mapsto xy$ is uniformly continuous in $\mathbb{R}^2$.
(c) The function $(x, y) \mapsto x + y$ is uniformly continuous in $[0, 1]^2$.
(d) The function $(x, y) \mapsto xy$ is uniformly continuous in $[0, 1]^2$.

3.7. Cauchy-Schwarz. Let V be a vector space over $\mathbb{R}$, let $\langle \cdot, \cdot \rangle$ be an inner product on V , and let $\| \cdot \| : V \rightarrow \mathbb{R}$ be given by $\|v\| = \sqrt{\langle v, v \rangle}$. Then the inequality

$$|\langle v, w \rangle| \leq \|v\| \|w\| \tag{1}$$

holds for all $v, w \in V$. Furthermore, equality in (1) holds if and only if v and w are linearly dependent.

3.8. Inner product and norm. Let V be a vector space over $\mathbb{R}$, let $\langle -, - \rangle$ be an inner product on V . The map defined by

$$\| \cdot \| : V \rightarrow \mathbb{R}, \quad \|v\| = \sqrt{\langle v, v \rangle}$$

satisfies the triangular inequality and is a norm.

3.9. All norms are equivalent in $\mathbb{R}^n$. Let $| \cdot |$ denote the standard Euclidean norm in $\mathbb{R}^n$ and let $f: \mathbb{R}^n \rightarrow [0, \infty)$ be another norm (that is, a function satisfying the properties of Definition 9.89).

1. Expressing x in a basis and using the “abstract” properties that f must have, show that there is a constant $C_1 > 0$ such that

$$f(x) \leq C_1 |x| \text{ for all } x \in \mathbb{R}^n.$$

2. Show that f is continuous in $\mathbb{R}^n$ (with respect to the standard distance of $\mathbb{R}^n$!).
3. Show that there is a number $c_2 > 0$ such that

$$f(x) \geq c_2 \text{ for all } |x| = 1.$$

4. Conclude that $f(x) \geq c_2|x|$ for all $x \in \mathbb{R}^n$.
5. Show that if $\tilde{f}$ is yet another norm, then there is $C > 0$ such that

$$C^{-1}f(x) \leq \tilde{f}(x) \leq Cf(x) \text{ for all } x \in \mathbb{R}^n.$$

3.10. Hilbert Schmidt norm of the composition. Take two linear functions $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^n$ and $\psi: \mathbb{R}^n \rightarrow \mathbb{R}^m$, and denote with Φ, Ψ the matrices that represent them in the canonical basis'. Recall that the linear map $\psi \circ \phi: \mathbb{R}^d \rightarrow \mathbb{R}^m$ is represented in these basis' by the matrix $\Psi \cdot \Phi$. Show that

$$\|\Psi \cdot \Phi\| \leq \|\Psi\| \|\Phi\|,$$

where $\|\cdot\|$ is the Hilbert-Schmidt norm of a matrix (see Definition 9.96 in the notes).

3.11. (*) Uniform continuity and moduli of continuity. Let (X, d_X) and (Y, d_Y) be metric spaces and let $f: X \rightarrow Y$.

A *modulus of continuity* is a function $\omega: [0, \infty) \rightarrow [0, \infty]$ such that

$$\omega(0) = 0, \quad \omega \text{ is nondecreasing}, \quad \lim_{t \downarrow 0} \omega(t) = 0.$$

(Here $[0, \infty]$ means we also allow the value $+\infty$.)

We say that f has *modulus of continuity* ω if

$$d_Y(f(x), f(y)) \leq \omega(d_X(x, y)) \quad \forall x, y \in X.$$

1. Prove that f is uniformly continuous on X if and only if f has a modulus of continuity.
2. (Bonus 1) Assume $X \subset \mathbb{R}$ is an interval with the usual distance $d_X(x, y) = |x - y|$. Show that the modulus constructed in (1) can be chosen *subadditive*, i.e.

$$\omega(t_1 + t_2) \leq \omega(t_1) + \omega(t_2) \quad \forall t_1, t_2 \geq 0.$$

3. (Bonus 2) Assume $X \subset \mathbb{R}^n$ is *convex*, meaning: for all $x_1, x_2 \in X$ and all $\lambda \in [0, 1]$,

$$(1 - \lambda)x_1 + \lambda x_2 \in X.$$

With $d_X(x, y) = \|x - y\|$ (Euclidean distance), show again that the modulus from (1) can be chosen subadditive.

Remark: In general metric spaces a modulus of continuity may have infinite value for finite $t > 0$. Example: $X = \mathbb{Z}$ with the usual distance, $Y = \mathbb{R}$, $f(n) = n^2$. This f is uniformly continuous on $\mathbb{Z}$ (because points closer than 1 must be equal), but no finite $\omega(1)$ can bound $|f(n+1) - f(n)| = 2n+1$ for all n . Allowing the value $+\infty$ (as above) avoids this issue, and in many familiar settings (e.g. intervals/convex sets in $\mathbb{R}^n$) the resulting $\omega_f(t)$ is finite for all t .

Hints:

- 3.2 If a function is continuous at 0 and $x_k \rightarrow 0$ and $y_k \rightarrow 0$, then $f(x_k)$ and $f(y_k)$ must be converging to the same value...
- 3.9.2 Recall that from the definition it follows that $|f(x) - f(y)| \leq f(x - y)$... And that Lipschitz functions are always continuous.
- 3.9.3 Apply Weierstrass' Theorem to f on $S := \{x \in \mathbb{R}^n : |x| = 1\}$ and use that f , being a norm, is nondegenerate.
- 3.9.5 If f is equivalent to $|\cdot|$ and $\tilde{f}$ is equivalent to $|\cdot|$ it follows by transitivity that f is equivalent to $\tilde{f}$...
- 3.10 First recall that $\|M\|^2$ is the sum of the squares of the entries of M . Notice that $(\Psi \cdot \Phi)_j^i$ (i th row and j th column) is the scalar product of Ψ^i and Φ_j which are vectors of $\mathbb{R}^n$. Apply Cauchy-Schwarz to each of them.
- 3.11.1 Define the “worst oscillation at scale t ”
- $$\omega_f(t) := \sup \left\{ d_Y(f(x), f(y)) : d_X(x, y) \leq t \right\}.$$
- 3.11.2 If $d_X(x, y) \leq t_1 + t_2$, choose an intermediate point z on the segment from x to y so that $d_X(x, z) \leq t_1$ and $d_X(z, y) \leq t_2$, then use the triangle inequality in Y .
- 3.11.3 If $d_X(x, y) \leq t_1 + t_2$, choose an intermediate point z on the segment from x to y so that $d_X(x, z) \leq t_1$ and $d_X(z, y) \leq t_2$, then use the triangle inequality in Y .

3. Solutions

Solution of 3.1: (a) $f(x) := \frac{1+x}{2}$. (b) $f(x, y) := (x + 1, y)$.

Solution of 3.2: Assume such continuous f exists, then it must be sequentially continuous. Lets us prove that this is never the case exhibiting two sequences converging to the origin along which f takes completely different values. Pick any sequence $t_k \downarrow 0$, then:

$$(t_k, t_k) \rightarrow (0, 0) \text{ and } (t_k, -t_k) \rightarrow (0, 0), \text{ but } \lim_k f(t_k, t_k) = \frac{1}{2} \neq -\frac{1}{2} = \lim_k f(t_k, -t_k).$$

First of all let us show that if such g exists it is necessarily given by

$$g(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } x = y = 0. \end{cases} \quad (2)$$

In fact $\lim_k g(1/k, 1/k) = \lim_k 1/k = 0$, so this can be the only value g takes at the origin.

Let us now check that the g defined by (2) is indeed continuous at the origin (continuity at the points outside the origin is obvious). Take any nonzero sequences $x_k, y_k \rightarrow 0$, then

$$|g(x_k, y_k)| = \frac{|x_k y_k|}{\sqrt{x_k^2 + y_k^2}} \leq \frac{x_k^2 + y_k^2}{\sqrt{x_k^2 + y_k^2}} \leq \sqrt{x_k^2 + y_k^2} \rightarrow 0.$$

Solution of 3.3:

1. Not open, since $(0, 2+\epsilon^2, 0)$ lies outside E_1 , but $(0, 2-\epsilon^2, 0)$ lies inside E_1 . Not closed, since $(0, \epsilon^2, 0)$ lies inside E_1 , but the limit point as $\epsilon \rightarrow 0$ does not: $(0, 0, 0) \notin E_1$. Not compact otherwise would be closed.
2. Notice that $E_2 = f^{-1}([1/4, \infty))$ with $f: \mathbb{R}^n \rightarrow \mathbb{R}$ give by $f(x) = \sin \|x\|$. Now f is continuous, $[1/4, \infty)$ is closed in $\mathbb{R}$, so E_2 is closed. On the other hand $\mathbb{R}^n$ is connected, so E_2 cannot be also open since:

$$(\pi/2, 0, \dots, 0) \in E_2, \quad (0, \dots, 0) \notin E_2.$$

Finally, E_2 cannot be compact because it is not bounded as $x_k := (2k\pi + \pi/2, 0, \dots, 0) \in E_2$ for all $k \in \mathbb{N}$.

3. Define the continuous functions $g, f_n: \mathbb{R}^3 \rightarrow \mathbb{R}$ as

$$f_n(x) := x_1^4 - \frac{1}{n}x_3^2 - x_2^2 - \frac{1}{n}, g(x) := 6 - x_1 - x_2.$$

Then we can write

$$E_3 := \bigcup_{n \geq 1} f_n^{-1}((-\infty, 0)) \cap g^{-1}((-\infty, 0)) = \left\{ \bigcup_{n \geq 1} f_n^{-1}((-\infty, 0)) \right\} \cap g^{-1}((-\infty, 0)),$$

the term in curly braces is open (union of open) and so is E_3 (we intersect only once with another open set). Notice that $(1+t, -t, 0) \in E_3$ for all $t \geq 1$ so E_3 is unbounded and it cannot be compact (and it is nonempty). On the other hand $(0, 6, 1) \notin E_3$, so its complementary cannot be empty, hence E_3 is not closed ($\mathbb{R}^n$ is connected again).

4. Similarly, open using the continuous function

$$(x, y) \mapsto x \cdot y - \|x\|\|y\|,$$

unbounded by homogeneity $(x, y) \in E_4 \Leftrightarrow (tx, ty) \in E_4$. Not empty as $(e_n, e_n) \in E_4$, not everything as $(0, v) \notin E_4$.

5. Invertibility can be expressed as $E_5 = \det^{-1}(\mathbb{R} \setminus \{0\})$, since the determinant is a polynomial function of the entries it is continuous, so E_5 is open. Again is unbounded by homogeneity and not everything (not all matrices are invertible).
6. $E_6 = f^{-1}(\{0\})$ with $f(X) = \sum_{i,j} |X_{ij} - X_{ji}|^2$. So E_6 is closed. Cannot be open (not all matrices are symmetric, there is at least one symmetric matrix) nor compact (one can find symmetric matrices with arbitrarily large entries).
7. This is a finite set of “points” in $\mathbb{R}^{n \times n}$. Hence it is compact, closed and not open.
8. $E_8 = f^{-1}(-\infty, 6]$ with $f(x) = \max_{1 \leq i \leq n} |x_i|$ so it is closed. It is not everything nor nothing, so it is not open. Furthermore it is compact since it is clearly bounded:

$$x \in E_8 \Rightarrow \|x\|^2 \leq \sum_{i=1}^n 6^2 \leq 100^n.$$

9. Since E_8 is unbounded so is E_9 , so it cannot be compact. Being a product of an open and a compact set we expect it to be none of them. If it was open, then write it as union of open balls $\{B(x_i, r_i)\}_{i \in I} \subset \mathbb{R}^{n+3}$. Then the balls

$$B' := B(x_i, r_i) \cap \{x_{n+1} = \dots = x_{n+3} = 0\} \subset \mathbb{R}^n$$

would be open and contained in E_8 , so E_8 would be open, contradiction.

Pick a sequence $\{x_k\} \subset E_3 \subset \mathbb{R}^3$ that converges to some $z \in \mathbb{R}^3 \setminus E_3$. Then the sequence $x'_k := (0, \dots, 0, x_k) \in \mathbb{R}^{n+3}$ is convergent to $(0, \dots, 0, z)$. If E_9 were closed we would get $(0, \dots, 0, z) \in E_8 \times E_3$, hence $z \in E_3$, contradiction.

10. Being an intersection of a closed and a compact set it is compact and so closed. It is not empty as $(\pi/2, 0, \dots, 0) \in E_{10}$, so it is not open.

Solution of 3.4:

- (a) True. Let U be such a set, then pick any $z \in \partial U$ and let $f \in C(U)$ be defined by $f(x) := \|x - z\|^{-1}$. If U was compact f would be bounded, we can pick a sequence $x_k \in U$ such that $x_k \rightarrow z$, so $f(x_k) \rightarrow \infty$, contradiction.

We remark that ∂U is not empty otherwise $U = \overline{U}$ would be clopen, but it is not empty nor the whole $\mathbb{R}^n$ by assumption.

- (b) True. Let U be such a set, pick $z \in \partial U$ and a sequence $x_k \in U$ such that $x_k \rightarrow z$. While $\{x_k\}$ is Cauchy (is convergent in $\mathbb{R}^n$) its limit point does not belong to U .
- (c) True, it must be closed as subset of $\mathbb{R}^n$.

- (d) False. A finite subset of $\mathbb{R}^n$ is always compact, thus complete, but $\mathbb{Q}$ is not complete, while being a countable union of points.
- (e) True. A sequence in $\mathbb{R}^n$ converges if and only if it is Cauchy, so, for a set, containing its accumulation points or the limits of its Cauchy sequences is the same property.

Solution of 3.5:

- (a) True, at least if your desk is in Zurich. The map (literally) maps Zurich to a piece of paper, so point which in reality are hundreds of meters apart are distant few millimeters in the map. So the Lipschitz constant is approximately $1/10000$. Furthermore your desk is in Zurich, so the map sends the domain in itself (if you are in e-learning from Singapore then it is false).
- (b) False, $f(x) := 2$ is a counterexample.
- (c) True, Banach's fixed point theorem applies since $[0, 1] \subset [0, 2]$.
- (d) False, take f that swaps $[0, 1] \leftrightarrow [2, 3]$ contracting them of a factor $1/10$. In symbols:

$$f(x) := 2 + \frac{x}{10} \text{ for all } x \in [0, 1], \quad f(x) := \frac{x-2}{10} \text{ for all } x \in [2, 3].$$

- (e) True. Take $\epsilon > 0$ and consider $f_\epsilon(x) := (1 - \epsilon)f(x)$. Notice that $f_\epsilon([0, 1]) \subset [0, 1 - \epsilon] \subset [0, 1]$. Also, $|f'_\epsilon(x)| < 1 - \epsilon$, so by the Mean Value Theorem it is a contraction (the interval is connected). So Banach's fixed point theorem gives $x_\epsilon \in [0, 1]$ such that

$$x_\epsilon = f_\epsilon(x_\epsilon) = (1 - \epsilon)f(x_\epsilon).$$

By compactness, there is $\epsilon_k \rightarrow 0$ and $x_0 \in [0, 1]$ such that $x_{\epsilon_k} \rightarrow x_0$, so we can pass this equation to the limit and find that x_0 is a fixed point.

Actually, applying the intermediate value theorem to the function $g(x) - x$, one can show that every continuous function $g: [0, 1] \rightarrow [0, 1]$ has a fixed point.

Solution of 3.6:

- (a) True, it is in fact Lipschitz continuous:

$$|(x + y) - (x' + y')| \leq |x - x'| + |y - y'| \leq 2\sqrt{|x - x'|^2 + |y - y'|^2},$$

where we used $a + b \leq 2\sqrt{a^2 + b^2}$.

- (b) False. Let $f(x, y) := xy$. Consider for some $\delta > 0$ fixed, the balls $B_t := B((t, t), \delta)$ centered at the point (t, t) . The size of the interval $f(B_t)$ must go to infinity as $t \rightarrow \infty$, so it is impossible that it stays contained in $[-1 + t^2, t^2 + 1] = B_1(f(t, t)) \subset \mathbb{R}$, violating the definition of uniform continuity with $\epsilon = 1$.

Check: $f(t, t \pm \delta) = t^2 \pm 2t\delta + \delta^2$, so $f(B_t) \subset \mathbb{R}$ is an interval (it is connected) that contains the interval $[t^2 - 2t\delta + \delta^2, t^2 + 2t\delta + \delta^2]$, whose length goes to infinity as $t \rightarrow \infty$ with δ fixed.

- (c) True because of the Heine-Cantor Theorem.
(d) True because of the Heine-Cantor Theorem.

Solution of 3.7: This is Proposition 9.100 in the notes. An alternative proof is:

We assume without loss of generality that $\|v\| \neq 0$ and $\|w\| \neq 0$, as otherwise the result is trivial.

Let $\lambda \in \mathbb{R}$. By definiteness, $\langle \lambda v + w, \lambda v + w \rangle \geq 0$; by bilinearity and symmetry, we can write this as

$$\lambda^2 \langle v, v \rangle + \lambda 2 \langle v, w \rangle + \langle w, w \rangle \geq 0.$$

Fixed v and w , this is just a 2nd order polynomial in the variable $\lambda \in \mathbb{R}$. Moreover, we know that it is a nonnegative function.

By the formula for the roots of a second order polynomial, we deduce that

$$(2 \langle v, w \rangle)^2 - 4 \langle v, v \rangle \langle w, w \rangle \leq 0,$$

which taking square roots gives

$$|\langle v, w \rangle| \leq \sqrt{\langle v, v \rangle} \sqrt{\langle w, w \rangle} = \|v\| \|w\|.$$

Finally, equality implies that the polynomial above has a zero (again, by the formula for the roots of a second order polynomial; in fact, it is given by $\lambda = \frac{-2 \langle v, w \rangle}{2 \langle v, v \rangle} = -\frac{\langle v, w \rangle}{\langle v, v \rangle}$). But this means that $\langle \lambda v + w, \lambda v + w \rangle \geq 0$ is zero for this value of λ , so that (by definiteness) $\lambda v + w = 0$, showing the desired linear dependence.

Solution of 3.8: This is Corollary 9.101 in the notes.

Solution of 3.9: This is Theorem 9.107 in the notes.

Solution of 3.10: For $i \in \{1, \dots, m\}$ we denote with Ψ^i the vector $(\Psi_\ell^i)_{1 \leq \ell \leq n}$ in $\mathbb{R}^n$. For $j \in \{1, \dots, d\}$ we denote with Φ_j the vector $(\Phi_j^\ell)_{1 \leq \ell \leq n}$ in $\mathbb{R}^n$. Notice that by definition of matrix multiplication

$$(\Psi \cdot \Phi)_j^i = \Psi^i \cdot \Phi_j.$$

Now, by Cauchy-Schwarz inequality in $\mathbb{R}^n$ we have

$$\|\Psi \cdot \Phi\|^2 = \sum_{i=1}^m \sum_{j=1}^d |\Psi^i \cdot \Phi_j|^2 \leq \sum_{i=1}^m \sum_{j=1}^d |\Psi^i|^2 |\Phi_j|^2 = \left(\sum_{i=1}^m |\Psi^i|^2 \right) \left(\sum_{j=1}^d |\Phi_j|^2 \right) = \|\Psi\|^2 \|\Phi\|^2.$$

Solution of 3.11: (1) $\Rightarrow$ (modulus). Assume f is uniformly continuous. Define for $t \geq 0$

$$\omega_f(t) := \sup \left\{ d_Y(f(x), f(y)) : d_X(x, y) \leq t \right\} \in [0, \infty].$$

Then for all $x, y \in X$ we have, by definition of supremum,

$$d_Y(f(x), f(y)) \leq \omega_f(d_X(x, y)).$$

Also ω_f is nondecreasing, and $\omega_f(0) = 0$.

It remains to show $\lim_{t \downarrow 0} \omega_f(t) = 0$. Let $\varepsilon > 0$. By uniform continuity there exists $\delta > 0$ such that $d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon$. If $0 < t < \delta$, then for every pair with $d_X(x, y) \leq t$ we have $d_X(x, y) < \delta$, hence $d_Y(f(x), f(y)) < \varepsilon$. Taking the supremum over all such pairs gives $\omega_f(t) \leq \varepsilon$. Therefore $\omega_f(t) \rightarrow 0$ as $t \downarrow 0$, so ω_f is a modulus of continuity for f .

(modulus) $\Rightarrow$ (1). Assume there exists a modulus ω with $d_Y(f(x), f(y)) \leq \omega(d_X(x, y))$ for all x, y . Let $\varepsilon > 0$. Since $\omega(t) \rightarrow 0$ as $t \downarrow 0$, choose $\delta > 0$ with $\omega(\delta) < \varepsilon$. If $d_X(x, y) < \delta$, then (using that ω is nondecreasing)

$$d_Y(f(x), f(y)) \leq \omega(d_X(x, y)) \leq \omega(\delta) < \varepsilon.$$

Thus f is uniformly continuous.

Bonus 1 (intervals in $\mathbb{R}$). Let X be an interval in $\mathbb{R}$ and take $\omega = \omega_f$ as above. Fix $t_1, t_2 \geq 0$ and $x, y \in X$ with $|x - y| \leq t_1 + t_2$. Assume $x \leq y$. If $y - x \leq t_1$, then $d_Y(f(x), f(y)) \leq \omega_f(t_1) \leq \omega_f(t_1) + \omega_f(t_2)$. Otherwise set $z := x + t_1$. Since X is an interval and $x \leq z \leq y$, we have $z \in X$. Moreover $|x - z| = t_1$ and

$$|z - y| = y - (x + t_1) = (y - x) - t_1 \leq t_2.$$

By the triangle inequality in Y ,

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(z)) + d_Y(f(z), f(y)) \leq \omega_f(t_1) + \omega_f(t_2).$$

Taking the supremum over all x, y with $|x - y| \leq t_1 + t_2$ gives

$$\omega_f(t_1 + t_2) \leq \omega_f(t_1) + \omega_f(t_2),$$

so ω_f is subadditive.

Bonus 2 (convex sets in $\mathbb{R}^n$). Let $X \subset \mathbb{R}^n$ be convex and again take $\omega = \omega_f$. Fix $t_1, t_2 \geq 0$ and $x, y \in X$ with $\|x - y\| \leq t_1 + t_2$. If $\|x - y\| \leq t_1$, we are done as above.

Otherwise define $\lambda := t_1 / \|x - y\| \in (0, 1)$ and set

$$z := (1 - \lambda)x + \lambda y.$$

By convexity, $z \in X$. Also

$$\|x - z\| = \|\lambda(y - x)\| = \lambda\|x - y\| = t_1, \quad \|z - y\| = \|(1 - \lambda)(y - x)\| = (1 - \lambda)\|x - y\| = \|x - y\| - t_1 \leq t_2.$$

Thus, using the triangle inequality in Y ,

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(z)) + d_Y(f(z), f(y)) \leq \omega_f(t_1) + \omega_f(t_2).$$

Taking the supremum over all x, y with $\|x - y\| \leq t_1 + t_2$ yields $\omega_f(t_1 + t_2) \leq \omega_f(t_1) + \omega_f(t_2)$, so ω_f is subadditive.